Q. 1 (c) 4% decrease

Assume the original number = 100.

- After 20 increase = 100 + 20 = 120
- After 20 decrease = 120 − 24 = 96

Net change = 100 − 96 = 4%.

final answer :- 4% decrease.

Q. 2. (d) 20%

formula

$$\text{Reduction \%} = \frac{x}{100+x} \times 100$$

Here, $x = 25\%$

$$= \frac{25}{125} \times 100$$

$$= 20\%$$

final answer : 20%

Q.3 (C) 25%

Assume CP of 1 article = ₹1

CP of 20 articles = ₹20.

According to the question:

SP of 16 articles = ₹20.

$$SP = \frac{20}{16} = ₹1.25$$

Profit = 1.25 − 1 = ₹0.25.

$$\text{Profit\%} = \frac{0.25}{1} \times 100 = 25\%$$

Final answer: 25%

Q.4. (a) 26%

Assume CP = ₹100.

marked price:

$$MP = 100 + 40 = ₹140$$

Discount = 10% of 140 = ₹14

Selling price:

$$SP = 140 - 14 = ₹126$$

Profit = 126 − 100 = ₹26.

$$\text{Profit \%} = 26\%$$

Final answer: 26%

Q.5. (d) 12.5 %

Assume Principal = ₹100

If money doubles, Amount = ₹200

Simple interest:

$$SI = 200 - 100 = ₹100$$

Formula:

$$SI = \frac{P \times R \times T}{100}$$

$$100 = \frac{100 \times R \times 8}{100}$$

$$R = \frac{100}{8} = 12.5\%$$

Final answer: 12.5 %

Q.6 (b) ₹2,100

formula:

$$A = P\left(1 + \frac{R}{100}\right)^T$$

$$A = 10000(1.10)^2$$

$$A = 10000 \times 1.21 = ₹12,100$$

Compound interest

$$CI = A - P$$

Final answer: ₹2,100 ₹2100 − 10000 = ₹2,100

Q.7 (a) ₹32

for 2 years:

$$\text{Difference} = P \times \left(\frac{R}{100}\right)^2$$

$$= 500 \times \left(\frac{8}{100}\right)^2$$

$$= 5000 \times 0.0064$$

$$= 32$$

Final answer: ₹32

Q.8 (b) 8:12:15

Given: A:B = 2:3

B:C = 4:5

Make B equal in both ratios

LCM of 3 and 4 = 12.

multiply first ration by 4:

A:B = 8:12

muliply second ration by 3:

B:C = 12:15

∴ A:B.C = 8:12:15

Final answer: 8:12:15.

Q9 (a) ₹5,400

Profit-sharing ratio = Investment × Time

A: $6000 \times 12 = 72000$

B: $8000 \times 6 = 48000$

$72000 : 48000 = 3 : 2$

total parts = $3 + 2 = 5$.

A's share:

$= \frac{3}{5} \times 9000$

$= 5,400$

Final answer: 5,400

Q10. (a) 28

Sum of ~~5~~ numbers. $5 \times 20 = 100$

Sum of remaining 4 numbers: $4 \times 18 = 72$

Removed numbers $100 - 72 = 28$

Final answer 28.

Q.11. (a) 46 years

total age of 30 students: $30 \times 15 = 450$

total age including teacher: $31 \times 16 = 496$

teacher's age: $496 - 450 = 46$

final answer: 46 years.

Q.12 (c) 6 days.

A's one-day work: $1/10$

B's one-day work: $1/15$

together $\frac{1}{10} + \frac{1}{15}$

LCM = 30

$$= \frac{3+2}{30} = \frac{5}{30} = \frac{1}{6}$$

∴ they complete the work in 6 days.

final answer 6 days.

Q. 13 (b) 30 days

together one day work 1/12

A's one-day work: 1/20

B's one-day work. $= \frac{1}{12} - \frac{1}{20}$

LCM = 60

$= \frac{5-3}{60} = \frac{2}{60} = \frac{1}{30}$

Final answer: 30 days.

Q. 14. (d) 18 hours

A's Filling rate: 1/6

B's emptying rate: 1/9

Net Filling rate: $\frac{1}{6} - \frac{1}{9}$

$= \frac{3-2}{18}$

$= \frac{1}{18}$

Final answer: 18 hours.

Q.15 (a) 10 seconds

convert km/h into m/s:

$54 \times 5/18 = 15$ m/s

formula

Time = Distance/speed

$= \frac{150}{15} = 10$

final answer: 10 seconds.

Q.16 (b) 25 seconds

total distance:

$200 + 300 = 500$ m

convert speed:

$72 \times \frac{5}{18} = 20$ m/s

Time

$= \frac{500}{20} = 25$

final answer: 25 seconds.

Q.17 (c) 10 hours

Downstream speed : 10 + 2 = 12 km/h

upstream speed : 10 - 2 = 8 km/h

Downstream time: 48/12 = 4 hours

upstream time: 48/8 = 6 hours

total time 4 + 6 = 10 hours

final answer: 10 hours.

Q.18 (d) 48 km/h

for equal distances:

$$\text{Average} = \frac{2xy}{x-y}$$

substitute

$$= \frac{2 \times 40 \times 60}{40 + 60}$$

$$= \frac{4800}{100} = 48$$

Final answer : 48 km/h.

Q.19 (b) 20 litres

total mixture = 60 litres.

milk: $2/3 \times 60 = 40L$

water $1/3 \times 60 = 20L$

for a 1:1 ratio, water must eqd milk
Required water = 40 litres.

Additional water:

$40 - 20 = 20L$

Final answer: 20 litres.

20. (c) 2:1

using the allegation method:

$$\text{Ratio} = \frac{55-45}{45-40}$$

$$= \frac{10}{5}$$

$$= 2:1$$

Final answer: 2:1.

Q.21 (d) 180

Formula:

$$\text{Product} = \text{HCF} \times \text{LCM}$$

$$2160 = 12 \times \text{LCM}$$

$$\text{LCM} = \frac{2160}{12} = 180$$

Final answer : 180

Q.22 (C) 32.4 litres

Where :

- P = 40 litres
- x = 4 litres
- V = 40 litres
- n = 2 times

Substitute $= 40\left(1 - \frac{4}{40}\right)^2$

$= 40(0.9)^2$

$= 40 \times 0.81$

$= 32.4\text{L}$

Final answer : 32.4 litres.

Q.23 (c) 30

the word LEVEL has 5 letters
- L occurs 2 times
- E occurs 2 times
- V occurs 1 time

formula: $n! / p! \times q!$

$= 5! / 2! \times 2!$

$= 120 / 2 \times 2$

$= 120 / 4 = 30$

final answer: 30 arrangements.

Q.24 (b) 74

total people: $5 + 4 = 9$

select any 3 people: ${}^{9}C_3 = \frac{9!}{3! \times 6!} = 84$

ways to select 3 men only ${}^{5}C_3 = 10$

required ways: $84 - 10 = 74$.

final answer: 74 ways.

Q. 25 (d) 5/36

Total possible outcomes: $6 \times 6 = 36$

Favourable outcomes for sum 8:

(2,6), (3,5), (4,4), (5,3), (6,2)

total favourable outcomes = 5.

Probability: $P = 5/36$

Final answer: 5/36.

Q. 26. (a) 2/15

total balls = 10.

Probability of first red ball 4/10

After taking one red ball:
Red balls reaining = 3
total balls reaiming = 9

Probability of second red ball: 3/9

Combined probability $= 4/10 \times 3/9$

$= 12/90 = 2/15$

Final answer: 2/15.

Q.27 (b) 66

total marks of section A: $40 \times 60 = 2400$

total marks of section B: $60 \times 70 = 4200$

combined marks $2400 + 4200 = 6600$

total students: $40 + 60 = 100$

Combined averge: $= 6600/100 = 66$

final answer: 66.

Q.28 (c) 700

Round the values:

- $49.98\% \approx 50\%$
- $801,2 \approx 800$
- $19.97 \approx 20$
- $15.02 \approx 15$

Now calculate: 50% of $800 = 400$

$20 \times 15 = 300$

Add

$400 + 300 = 700$

Final answer: Approximately 700.

Q.29 (d) 820

Given • first term (a) = 3
• common difference (d) = 7 − 3 = 4
• number of terms (n) = 20

Substitute $S_{20} = 20/2 [2(3) + (20-1)4]$

$= 10 [6 + 76]$

$= 10 \times 82$

$= 820$.

Final answer : 820

Q.30 (b) 24 years.

Let ~~Res~~ Present ages be:

$A = 4x$ After 6 years $\frac{4x+6}{5x+6} = \frac{5}{6}$

$B = 5x$

cross multiply : $6(4x+6) = 5(5x+6)$

$24x + 36 = 25x + 30$

$36 - 30 = 25x - 24x$

$x = 6$

A's present age

$4x = 4 \times 6 = 24$

Final answer : 24 years.